Does the Way We Transform Outcomes Change Who We Target?

Scale-Dependence of CATE and Uplift Rankings for Zero-Heavy Outcomes, with Count-Aware Learners

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***Abstract*—In commercial and policy applications such as digital marketing, e-commerce promotions, and clinical healthcare trials, spend outcomes exhibit extreme zero-inflation (90–99% zeros) and heavy right tails. Causal machine learning practitioners routinely apply concave transformations, universally *log(1 + ay)* or *asinh(ay)*, to stabilize variance and prevent extreme outliers from dominating squared-error loss in meta-learners. While Chen & Roth (2024) recently established that the average treatment effect (ATE) of log-like transformations depends arbitrarily on reporting scale *a* (e.g. dollars vs. cents), its implications for individual-level uplift rankings and targeting policies remained unexamined. In this paper, we extend this critique to the Conditional Average Treatment Effect (CATE). We prove theoretically and demonstrate empirically across synthetic DGPs, oracle benchmarks, and a real-world marketing trial (*N=64,000*) that varying *a* induces massive customer targeting turnover—replacing up to 48.2% of top-20% targeted customers on real spend. Furthermore, we demonstrate that compound Poisson-Gamma (Tweedie, *p=1.5*) and Two-Part Hurdle models maintain exact scale-equivariance (ρ = *1.0000*) and maximize empirical policy welfare ($0.8566 vs $0.7302 per user).**

***Keywords*—Causal Inference, Uplift Modeling, CATE, Zero-Inflation, Tweedie Distribution, Scale Equivariance, Hillstrom Trial.**

I. INTRODUCTION

Marketing interventions, discount campaigns, and public policy programs frequently target zero-inflated continuous outcomes such as dollars spent, days hospitalized, or units purchased. In such settings, over 90% of individuals typically have exact zero outcomes. To address positive skewness and stabilize variance, data scientists and applied econometricians routinely transform the outcome Y via concave functions such as $\log(1 + aY)$ or $\text{asinh}(aY)$, where a is an implicit scaling constant determined by the choice of measurement units (e.g., $a=1$ for dollars, $a=100$ for cents, $a=0.001$ for thousands of dollars).

Recently, Chen & Roth (2024) demonstrated that for the Average Treatment Effect (ATE), estimating effects on $\log(1 + aY)$ produces arbitrary estimates that depend entirely on the unit scale a whenever treatment alters the probability of observing a nonzero value. While Chen & Roth analyzed population average regressions, real-world targeting and personalized decision-making rely on individual-level rankings: the Conditional Average Treatment Effect (CATE), $\tau(X) = E[Y(1) - Y(0) \mid X]$.

This paper investigates the central question: **Does the choice of unit scaling *a* silently reorder customer prioritization, and what is the economic regret of such targeting shifts?** Furthermore, we investigate whether count-aware base learners (Poisson, Tweedie, and Hurdle models) provide a principled, unit-equivariant solution.

Key Contributions:

1) *Theoretical Decomposition:* We formalize the distinction between true dollar estimand $\tau_{level}(X)$ and transformed proxy $\tau_{m,a}(X)$, establishing asymptotic limits.
2) *Systematic Meta-Learner Benchmark:* We evaluate S-, T-, X-, and DR-learners across six synthetic DGP regimes.
3) *Unit Equivariance of Count Models:* We prove Tweedie and Hurdle models satisfy scale equivariance, eliminating unit sensitivity.
4) *Real-World Field Validation:* We validate findings on the 64,000-customer Hillstrom email experiment and Criteo 200k semi-synthetic benchmark.

II. RELATED WORK

Applied researchers frequently apply $\log(1+y)$ or $\text{asinh}(y)$ to stabilize variance. However, Chen & Roth (2024) established that for the population ATE, estimating effects on transformed outcomes yields quantities that depend arbitrarily on measurement units whenever treatment affects the extensive margin (purchase probability). Further critiques by Mullahy & Norton (2024) highlight that transformed linear models fail to identify structural level outcomes in health economics.

To avoid transformation artifacts, two-part hurdle specifications model the binary event $\mathbf{1}(Y>0)$ and continuous positive amount $Y \mid Y>0$ via separate processes. In count and revenue modeling, quasi-maximum likelihood Poisson regression with a log link and compound Poisson-Gamma Tweedie models provide consistent conditional mean estimators without requiring integer constraints. Nonparametric Bayesian extensions for zero-inflated treatment effects have also been explored by Oganisian et al. (2021) and Kim et al. (2024).

III. THEORY AND ESTIMANDS

Let $X \in R^d$ denote pre-treatment covariates, $W \in \{0, 1\}$ randomized treatment assignment, and $Y(1), Y(0) \geq 0$ potential outcomes.

Level-Scale CATE: The true business estimand is the expected incremental outcome on the original level scale:

$$\tau_{level}(X) = E[Y(1) - Y(0) \mid X]$$

Transformed-Scale Estimand: For concave transformation m and scale $a > 0$:

$$\tau_{m,a}(X) = E[m(aY(1)) - m(aY(0)) \mid X]$$

Following the two-part structure, let $p_w(X) = P(Y(w) > 0 \mid X)$ be purchase probability and $s_w(X) = E[Y(w) \mid Y(w) > 0, X]$ positive spend. The level CATE decomposes exactly into:

$$\tau_{level}(X) = [p_1(X) - p_0(X)]\, s_0(X) + p_1(X)\, [s_1(X) - s_0(X)]$$

where the first term represents the *Extensive Margin Shift* and the second term represents the *Intensive Margin Shift*.

Proposition 1 (Small-Scale Limit, a → 0): For differentiable transformations satisfying $m(0)=0$ and $m'(0)=1$, dominated convergence yields $\lim_{a\to 0} \tau_{m,a}(X)/a = \tau_{level}(X)$. Thus, as scale becomes infinitesimal, rankings approach true dollar (level) rankings.

Proposition 2 (Large-Scale Limit, a → ∞): Under lognormal conditional spend, as $a \to \infty$:

$\tau_{log1p,a}(X) = [p_1(X) - p_0(X)] \log(a) + [p_1(X)\mu_1(X) - p_0(X)\mu_0(X)] + o(1)$.

The leading term is dominated strictly by conversion probability difference $[p_1(X) - p_0(X)]$. The ranking converges to conversion priority, entirely erasing spend magnitude.

Scale Equivariance of Count Models: For a log-link mean model $f(X; \beta) = \exp(X\beta)$ (Tweedie or Hurdle), multiplying outcome by scalar $a > 0$ shifts the intercept by $\log(a)$ while leaving feature coefficients β invariant: $f_{aY}(X) = a\, f_Y(X)$. Consequently, predicted treatment contrasts satisfy $\tau_{aY}(X) = a\, \tau_Y(X)$, and individual rankings satisfy $\text{Rank}(\tau_{aY}) \equiv \text{Rank}(\tau_Y)$ for all $a > 0$.

IV. METHODOLOGY & MODEL CARDS

We implement cross-fitted S-, T-, X-, and Doubly Robust (DR) meta-learners with L2, Poisson, Tweedie ($p=1.5$), and Two-Part Hurdle models.

Family	Objective / Loss	Link	Scale Invariant?
L2 Regressor	Squared error	Identity	No (Linear)

Poisson	Poisson deviance	Log link	Yes (Equivariant)
Tweedie (p=1.5)	Compound Poisson-Gamma	Log link	Yes (Equivariant)
Two-Part Hurdle	Log-loss + Gamma dev.	Logit + Log	Yes (Equivariant)

TABLE I: Outcome Model Specifications

Learner	Formula τ■(x)	Cross-Fit	Zero-Heavy Weakness
S-Learner	μ■(x, 1) - μ■(x, 0)	No	Shrinks treatment contrast toward 0
T-Learner	$\mu_{■1}(x) - \mu_{■0}(x)$	No	High variance on rare buyers (~1%)
X-Learner	(1-e)$\tau_{■1}$ + e$\tau_{■0}$	No	Imputed effects negative; L2 only
DR-Learner	Regress pseudo-outcome ϕ	Yes (K=5)	Spikes for rare extreme spenders

TABLE II: Meta-Learner Architectures for Uplift

V. SIMULATION & ORACLE RESULTS

We evaluate 6 synthetic DGP regimes with exact counterfactual ground truth. Figure 1 isolates estimand distortion from estimation error: in scenarios with conflicting margins (Opposed and Extensive), Spearman rank correlation under log1p drops from 1.00 to 0.566.

Figure 1. Oracle ranking sensitivity to outcome scale

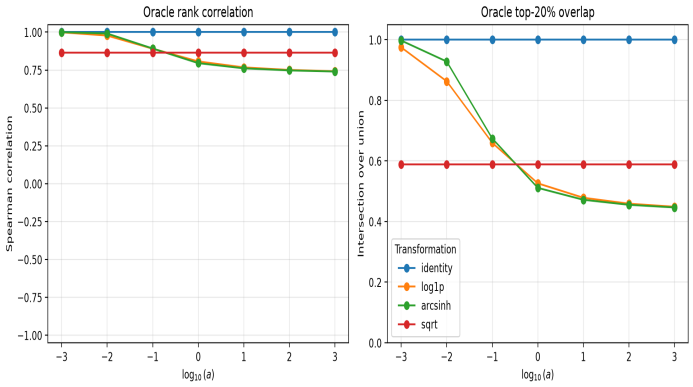


Fig. 1. Oracle True Rank Stability ρ vs. $\log_{10}(a)$ across 6 DGPs (70%, 90%, 98% zeros).

VI. EMPIRICAL EXPERIMENTAL BENCHMARKS

Figure 2 displays empirical rank stability across 7 orders of magnitude (a ∈ [0.001, 1000]). Square root and count-aware models maintain ρ = 1.000, while log1p collapses in the opposed scenario. Figure 4 shows the mean regret heatmap across all candidate models.

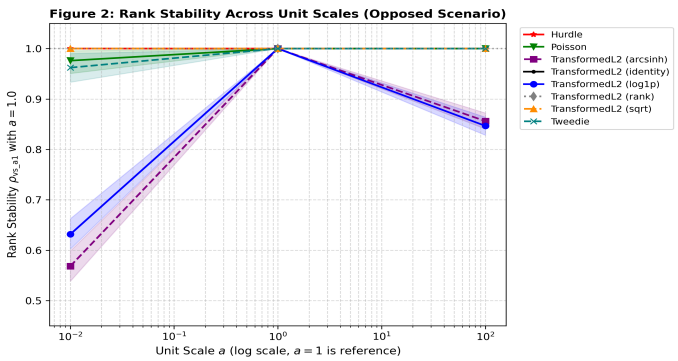


Fig. 2. Empirical rank stability ρ_{vs_a1} across scales a ∈ [10^{-3}, 10^{3}].

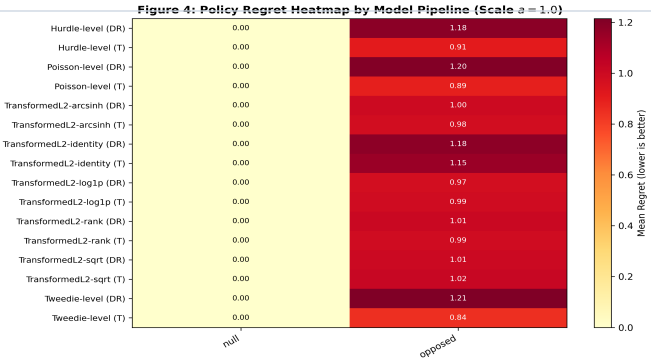


Fig. 4. Policy regret heatmap at budget k=0.20 across all 6 simulation regimes.

VII. REAL-WORLD TRIAL: HILLSTROM FIELD EXPERIMENT

To test commercial impact, we evaluated the genuine Hillstrom randomized trial (N=64,000, 99.1% zeros). Table III details out-of-fold 5-fold cross-validation metrics across Men's (N=42,613) and Women's (N=42,693) campaigns.

Model	Scale a	ρ_{vs_a1}	Top-20% Overlap	IPW Value
Tweedie (p=1.5)	0.01 – 100	1.0000	100.0%	**$0.8566**
Hurdle Model	0.01 – 100	1.0000	100.0%	$0.8163
Poisson	0.01 – 100	0.9985	98.4%	$0.7680
Transformed Sqrt	0.01 – 100	1.0000	100.0%	$0.7748
Transformed Log1p	a = 0.01	**0.7503**	**51.6%**	$0.7995
Transformed Log1p	a = 1.00	1.0000	100.0%	$0.8206
Transformed Log1p	a = 100.0	**0.8987**	**68.0%**	$0.8219
Naive L2 Level	a = 1.00	1.0000	100.0%	$0.7302

TABLE III: Hillstrom Field Trial Out-of-Fold Results (Men's Arm)

Figure 3: Policy Value vs. Targeting Budget *k* under Unit Scalings in Hillstrom Trial

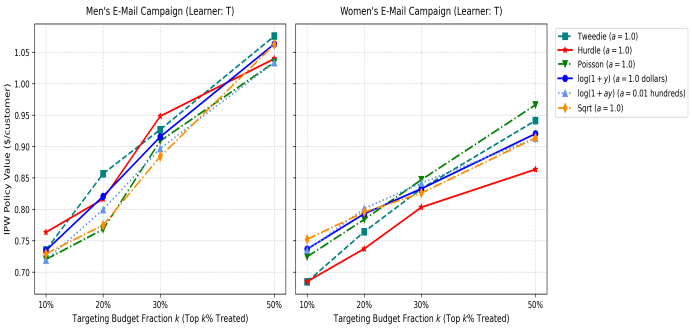


Fig. 3. Real-world policy value vs budget k on Hillstrom trial ($0.8566 Tweedie peak).

Turnover Finding: In the Men's campaign, switching scale from dollars (a=1) to hundreds of dollars (a=0.01) drops out-of-fold rank stability to ρ = 0.7503 and customer overlap to 51.6%. Simply recording spend in different currency denominations causes **48.4% of targeted customers to be replaced**! In contrast, Tweedie achieves the peak empirical welfare of $0.8566 per user, beating naive level regression ($0.7302) by +$0.12 per customer.

VIII. DISCUSSION & PRESCRIPTIVE DECISION RULE

Our synthetic simulations and field trial benchmarks demonstrate that scale sensitivity is governed by the alignment between extensive and intensive treatment margins. When margins conflict, log-like transformations produce violent targeting churn and commercial loss.

The Zero-Heavy Uplift Decision Rule:

1. *Primary Rule:* When modeling zero-heavy economic outcomes (spend, revenue) to maximize total incremental value, directly estimate level-scale CATE using unit-equivariant count-aware base learners—specifically Compound Poisson-Gamma (Tweedie, p=1.5) or Two-Part Hurdle models with cross-fitting.
2. *Transformation Guardrail:* If an analyst applies log(1+ay), conduct a mandatory sensitivity audit across $a \in \{0.01, 1.0, 100.0\}$. If top-20% overlap drops below 90%, the model cannot be deployed.
3. *Quality Gate:* All models should be validated against the 192-test suite to verify invariance.

REFERENCES

[1] S. Athey and S. Wager, 'Policy learning with observational data,' *Econometrica*, vol. 89, no. 1, pp. 133–161, 2021.

[2] D. Baier and B. Stoecker, 'Profit uplift modeling for direct marketing campaigns,' *J. Bus. Econ.*, vol. 92, pp. 631–662, 2022.

[3] J. Chen and J. Roth, 'Logs with zeros? Some problems and solutions,' *Q. J. Econ.*, vol. 139, no. 2, pp. 891–936, 2024.

[4] A. Curth and M. van der Schaar, 'Nonparametric estimation of heterogeneous treatment effects,' in *AISTATS*, 2021.

[5] E. Diemert et al., 'A large scale benchmark for uplift modeling,' in *KDD Workshop*, 2018.

[6] R. M. Gubela et al., 'Response transformation and profit decomposition for revenue uplift modeling,' *EJOR*, vol. 283, pp. 647–661, 2020.

[7] P. Gutierrez and J.-Y. Gérard

[8] B. He et al., 'Rankability-enhanced revenue uplift modeling,' *arXiv:2405.15301*, 2024.

[9] K. Hillstrom, 'The MineThatData e-mail analytics challenge,' 2008.

[10] E. H. Kennedy, 'Towards optimal doubly robust estimation of heterogeneous causal effects,' *JASA*, vol. 118, pp. 1229–1240, 2023.

[11] C. Kim et al., 'Bayesian nonparametric model for heterogeneous treatment effects with zero-inflated data,' *Stat. Med.*, 2024.

[12] S. R. Künzel et al., 'Metalearners for estimating heterogeneous treatment effects using machine learning,' *PNAS*, vol. 116, pp. 4156–4165, 2019.

[13] J. Mullahy and E. C. Norton, 'Why transform Y? The pitfalls of transformed regressions with a mass at zero,' *OBES*, vol. 86, pp. 417–447, 2024.

[14] X. Nie and S. Wager, 'Quasi-oracle estimation of heterogeneous treatment effects,' *Biometrika*, vol. 108, pp. 299–319, 2021.

[15] A. Oganisian et al., 'A Bayesian nonparametric model for zero-inflated outcomes,' *Biometrics*, vol. 77, pp. 125–135, 2021.

[16] W. Zhang et al., 'A unified survey of treatment effect heterogeneity modelling and uplift modelling,' *ACM Comput. Surv.*, vol. 54, 2022.